

Successive Differentiation

Definition: Successive Differentiation is the process of differentiating a given function successively times and the results of such differentiation are called successive derivatives. The higher order differential coefficients are of utmost importance in scientific and engineering applications.

Problem.1 i) If $y = x^3$, then find y_3 .

Solution: Here $y = x^3$

Differentiating this with respect to x ,

We get, $y_1 = \frac{dy}{dx} = 3x^2$, [Since $\frac{dx^n}{dx} = nx^{n-1}$]

Again, differentiating with respect to x ,

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (3x^2) = 3.2x = 6x$$

$$\text{Or, } y_2 = \frac{d^2y}{dx^2} = 6x$$

$$\text{Then } y_3 = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d}{dx} (6x) = 6.1 = 6$$

$$\therefore y_3 = \frac{d^3y}{dx^3} = 6. \quad \text{Ans. } y_3 = 6.$$

Problem.1 ii) If $y = x^n$, where n is a positive integer. Then find y_n .

Solution: Here $y = x^n$

Differentiating this with respect to x ,

We get, $y_1 = \frac{dy}{dx} = nx^{n-1}$; [Since $\frac{dx^n}{dx} = nx^{n-1}$]

$$y_2 = \frac{d^2y}{dx^2} = n(n-1)x^{n-2};$$

$$y_3 = \frac{d^3y}{dx^3} = n(n-1)(n-2)x^{n-3}; \text{ and proceeding in a similar manner.}$$

$$y_n = n(n-1)(n-2)(n-3) \dots \dots 3.2.1. x^{n-n}$$

$$y_n = n! x^0 = n! \quad \text{Answer. } y_n = n!$$

Problem.1 iii) If $y = e^{ax}$, then find y_n .

Solution: Here $y = e^{ax}$

Differentiating this with respect to x ,

We get, $y_1 = \frac{dy}{dx} = ae^{ax}$; [since $\frac{de^{ax}}{dx} = ae^{ax}$]

Again, differentiating with respect to x ,

$$y_2 = \frac{d^2y}{dx^2} = a^2 e^{ax};$$

$$y_3 = a^3 e^{ax}; \text{ and proceeding in a similar manner.}$$

$$y_n = a^n e^{ax}. \quad \text{Answer.}$$

HW. **Problem.1** iv) If $y = x^5$, then find y_3 . **Answer.** $y_3 = 60x^2$.

v) If $y = e^{mx}$, then find y_n . **Answer.** $y_n = m^n e^{mx}$.

Problem.2 Find y_2 for the following functions:

i) $y = e^{3x+2}$ ii) $x = a \cos \theta$, $y = a \sin \theta$. iii) $y = e^{5x+7}$.

Solution: i) Here $y = e^{3x+2}$ (1)

Differentiating this with respect to x ,

We get, $y_1 = \frac{dy}{dx} = \frac{d}{dx}(e^{3x+2}) = e^{3x+2} \frac{d}{dx}(3x+2)$; [since $\frac{de^{ax}}{dx} = ae^{ax}$]

$$\therefore y_1 = e^{3x+2}(3.1 + 0) = 3e^{3x+2}$$

Again, differentiating with respect to x ,

$$y_2 = 3 \frac{d}{dx}(e^{3x+2}) = 3e^{3x+2} \frac{d}{dx}(3x+2) = 3e^{3x+2}(3.1 + 0)$$

$$\therefore y_2 = 9e^{3x+2} = 9y \quad [\text{Using equation (1)}]$$

Thus $y_2 = 9y$ (**Answer**)

ii) Here $x = a \cos \theta$, $y = a \sin \theta$ (1)

Differentiating this with respect to θ ,

We get, $\frac{dx}{d\theta} = a \frac{d(\cos \theta)}{d\theta} = a(-\sin \theta) = -a \sin \theta$

$$\frac{dy}{d\theta} = a \frac{d(\sin \theta)}{d\theta} = a(\cos \theta) = a \cos \theta$$

$$y_1 = \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \cos \theta}{-a \sin \theta} = -\cot \theta$$

$$\therefore y_1 = -\cot \theta$$

Again, differentiating with respect to x ,

$$y_2 = \frac{d}{dx}(-\cot \theta) = -(-\operatorname{cosec}^2 \theta) \frac{d\theta}{dx} = \operatorname{cosec}^2 \theta \times \frac{1}{-a \sin \theta}$$

$$y_2 = -\frac{1}{a} \operatorname{cosec}^2 \theta \times \operatorname{cosec} \theta = -\frac{1}{a} \operatorname{cosec}^3 \theta \quad \text{Answer.}$$

HW.2 iii) $y = e^{5x+7}$ **Answer.** $y_2 = 25e^{5x+7} = 25y$.

Problem-3: If $y = 500e^{7x} + 600e^{-7x}$, Then show that $y_2 - 49y = 0$.

Solution: Given, $y = 500e^{7x} + 600e^{-7x}$ (1)

Differentiating this with respect to x ,

We get, $y_1 = \frac{dy}{dx} = 500 \frac{d}{dx}(e^{7x}) + 600 \frac{d}{dx}(e^{-7x})$

$$= 500(7e^{7x}) + 600(-7e^{-7x}) \quad [\text{since } \frac{de^{ax}}{dx} = ae^{ax}]$$

$$\therefore y_1 = 7(500e^{7x} - 600e^{-7x})$$

Again, differentiating with respect to x ,

$$y_2 = 7\{500(7e^{7x}) - 600(-7e^{-7x})\} = 49(500e^{7x} + 600e^{-7x})$$

$$\therefore y_2 = 49y \quad [\text{Using equation (1)}]$$

Thus $y_2 - 49y = 0$ Hence Shown

Problem-4: If $y = 2 + \log x$, Then show that $xy_2 + y_1 = 0$.

Solution: Given, $y = 2 + \log x$ (1)

Differentiating this with respect to x ,

$$\text{We get, } y_1 = \frac{dy}{dx} = 0 + \frac{1}{x} \quad [\text{since } \frac{d}{dx}(\log x) = \frac{1}{x}]$$

$$\therefore y_1 = \frac{1}{x}$$

Again, differentiating with respect to x ,

$$\text{So, } y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2} \quad [\text{Since } \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}]$$

$$\therefore y_2 = -\frac{1}{x^2}$$

$$\text{So, LHS} = xy_2 + y_1 = x\left(-\frac{1}{x^2}\right) + \frac{1}{x} = -\frac{1}{x} + \frac{1}{x} = 0 = \text{RHS}$$

Thus LHS = RHS (Hence Shown)

Problem-5: If $y = a \cos mx + b \sin mx$ Then show that $y_2 + m^2y = 0$.

Solution: Given, $y = a \cos mx + b \sin mx$ (1)

Differentiating this with respect to x ,

$$\text{We get, } y_1 = \frac{dy}{dx} = a(-m \sin mx) + b(m \cos mx)$$

$$\therefore y_1 = m(-a \sin mx + b \cos mx) \quad [\text{Since } \frac{d(\cos mx)}{dx} = -m \sin mx \text{ \& } \frac{d(\sin mx)}{dx} = m \cos mx]$$

Again, differentiating with respect to x ,

$$\text{So, } y_2 = m\{-a(m \cos mx) + b(-m \sin mx)\}$$

$$\text{or, } y_2 = -m^2(a \cos mx + b \sin mx)$$

$$\therefore y_2 + m^2y = 0 \quad [\text{Using equation (1)}]$$

Hence Shown

Problem-6: If $y = \sin(\log x)$, Then prove that $x^2y_2 + xy_1 + y = 0$.

Solution: Given, $y = \sin(\log x)$ (1)

Differentiating this with respect to x ,

$$\text{We get, } y_1 = \frac{dy}{dx} = \cos(\log x) \times \frac{d}{dx}(\log x) = \cos(\log x) \times \frac{1}{x} \quad [\text{Since } \frac{d}{dx}(\log x) = \frac{1}{x}]$$

$$\text{or, } y_1 = \frac{1}{x} \cos(\log x)$$

$$\therefore xy_1 = \cos(\log x) \dots \dots \dots (2)$$

Again, differentiating with respect to x ,

$$x \frac{dy_1}{dx} + y_1 \times \frac{dx}{dx} = -\sin(\log x) \frac{d}{dx}(\log x) \quad [\text{since } \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}]$$

$$\text{or, } xy_2 + y_1 \times 1 = -\sin(\log x) \times \frac{1}{x}$$

$$\text{or, } x^2 y_2 + xy_1 = -y$$

$$\therefore x^2 y_2 + xy_1 + y = 0. \text{ Hence Proved}$$

Problem-7: If $y = \tan x$, Then prove that $y_2 - 2yy_1 = 0$.

Solution: Given, $y = \tan x$ (1)

Differentiating this with respect to x ,

$$\text{We get, } y_1 = \frac{dy}{dx} = \sec^2 x \quad \left[\text{Since } \frac{d}{dx}(\tan x) = \sec^2 x \right]$$

$$\therefore y_1 = \sec^2 x \quad \dots \dots \dots (2)$$

Again, differentiating with respect to x ,

$$\text{So, } y_2 = 2 \sec x \times \frac{d}{dx}(\sec x) = 2 \sec x \times \sec x \tan x \quad \left[\text{since } \frac{d}{dx}(\sec x) = \sec x \tan x \right]$$

$$\text{or, } y_2 = 2 \sec^2 x \tan x$$

$$\text{or, } y_2 = 2yy_1 \quad [\text{Using equation (1) \& (2)}]$$

$$\therefore y_2 - 2yy_1 = 0 \quad \text{Hence Proved}$$

Problem-8: If $y = 2 \sin x + 3 \cos x$ Then show that $y_2 + y = 0$.

Solution: Given, $y = 2 \sin x + 3 \cos x$ (1)

Differentiating this with respect to x ,

$$\text{We get, } y_1 = \frac{dy}{dx} = 2 \cos x - 3 \sin x$$

Again, differentiating with respect to x ,

$$\text{So, } y_2 = 2(-\sin x) - 3 \cos x$$

$$\text{or, } y_2 = -2 \sin x - 3 \cos x \quad \dots \dots \dots (2)$$

Adding equations (2) & (1) we get,

$$y_2 + y = 2 \sin x + 3 \cos x - 2 \sin x - 3 \cos x = 0$$

$$\therefore y_2 + y = 0. \quad \text{Hence Showed}$$

Problem.9: If $y = a \cos(\log x) + b \sin(\log x)$, then show that $x^2 y_2 + xy_1 + y = 0$.

Solution: Given $y = a \cos(\log x) + b \sin(\log x)$<1>

Differentiating equation <1> two times with respect to x ,

$$\text{We get } y_1 = -a \sin(\log x) \cdot \frac{1}{x} + b \cos(\log x) \cdot \frac{1}{x}$$

Multiplying both sides by x

$$xy_1 = -a \sin(\log x) + b \cos(\log x)$$

$$\text{And } xy_2 + y_1 = -a \cos(\log x) \cdot \frac{1}{x} - b \sin(\log x) \cdot \frac{1}{x}$$

Multiplying both sides by x

$$x^2 y_2 + xy_1 = -\{a \cos(\log x) + b \sin(\log x)\} = -y \quad [\text{Using equation <1>}]$$

$$x^2 y_2 + xy_1 + y = 0 \quad \text{Hence showed.}$$

➤ Leibnitz's Theorem:

(n^{th} derivative of the product of two functions)

Statement: If u and v are two functions of x , each possessing derivatives up to n^{th} order, then the n^{th} derivative of the product, i.e.,

$$(uv)_n = u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + u v_n,$$

Where the suffixes of u and v denote the order of differentiations of u and v with respect to x .

Problem.10 If $y = e^{ax} x^3$, then find y_n .

Solution: Let $u = e^{ax}$, $v = x^3$

$$\begin{array}{l|l} \therefore u_1 = a e^{ax} & \text{and } v_1 = 3x^2 \\ \therefore u_2 = a^2 e^{ax} & \therefore v_2 = 3 \cdot 2 \cdot x = 6x \\ \therefore u_3 = a^3 e^{ax} & \therefore v_3 = 6 \\ \dots\dots\dots & \text{So, } v_4 = 0. \end{array}$$

$$\text{Thus, } u_n = a^n e^{ax}$$

Therefore, by Leibnitz's Theorem

$$(uv)_n = u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + u v_n \dots \dots \dots (1)$$

$$\text{Since } {}^n C_r = \frac{n!}{r! (n-r)!}; \quad {}^n C_1 = \frac{n!}{1! (n-1)!} = \frac{n(n-1)!}{1 \cdot (n-1)!} = n;$$

$${}^n C_2 = \frac{n!}{2! (n-2)!} = \frac{n(n-1)(n-2)!}{2 \cdot 1 \cdot (n-2)!} = \frac{n(n-1)}{2};$$

$${}^n C_3 = \frac{n!}{3! (n-3)!} = \frac{n(n-1)(n-2)(n-3)!}{3 \cdot 2 \cdot 1 \cdot (n-3)!} = \frac{n(n-1)(n-2)}{6}$$

Thus, from equation (1)

$$\begin{aligned} y_n &= a^n e^{ax} x^3 + n a^{n-1} e^{ax} \cdot 3x^2 + \frac{n(n-1)}{2} a^{n-2} e^{ax} \cdot 6x + \frac{n(n-1)(n-2)}{6} a^{n-3} e^{ax} \cdot 6 + 0 \\ \therefore y_n &= a^n x^3 e^{ax} + 3n a^{n-1} x^2 e^{ax} + 3n(n-1) a^{n-2} x e^{ax} + n(n-1)(n-2) a^{n-2} e^{ax}. \end{aligned}$$

(Answer)

Problem.11 Differentiating n times the equation $(1 + x^2)y_2 + (2x - 1)y_1$.

Solution: Given $(1 + x^2)y_2 + (2x - 1)y_1 \dots\dots\dots <1>$

For first part, $(1 + x^2)y_2$

Let $u = y_2$, $v = 1 + x^2$

$$\begin{array}{l|l} \therefore u_1 = y_3 & \text{and } v_1 = 0 + 2x \\ \therefore u_2 = y_4 & \therefore v_2 = 2 \\ \dots\dots\dots & \end{array}$$

$$\text{Thus, } u_n = y_{n+2}$$

Therefore, by Leibnitz's Theorem

$$(uv)_n = u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + u v_n$$

$$\{(1 + x^2)y_2\}_n = y_{n+2}(1 + x^2) + n \cdot y_{n+1} \cdot 2x + \frac{n(n-1)}{2!} y_n \cdot 2$$

$$\therefore \{(1+x^2)y_2\}_n = (1+x^2)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n$$

For 2nd part, $(2x-1)y_1$.

Let $u = y_1$, $v = 2x-1$

$$\begin{array}{l|l} \therefore u_1 = y_2 & \text{and } v_1 = 2.1 - 0 = 2 \\ \therefore u_2 = y_3 & \therefore v_2 = 0 \\ \dots\dots\dots & \\ \text{Thus, } u_n = y_{n+1} & \end{array}$$

Therefore, by Leibnitz's Theorem

$$\begin{aligned} \{(2x-1)y_1\}_n &= y_{n+1}(2x-1) + n \cdot y_n \cdot 2 + 0 \\ &= (2x-1)y_{n+1} + 2ny_n \end{aligned}$$

Adding these two

$$\begin{aligned} &(1+x^2)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n + (2x-1)y_{n+1} + 2ny_n \\ \therefore (1+x^2)y_{n+2} &+ \{2(n+1)x-1\}y_{n+1} + n(n+1)y_n \quad \text{(Answer)} \end{aligned}$$

Problem.12 If $y = \cos(10 \cos^{-1} x)$ show that $(1-x^2)y_{12} = 21xy_{11}$.

Solution: Given $y = \cos(10 \cos^{-1} x)$ <1>

Differentiating with respect to x we get,

$$\begin{aligned} y_1 &= -\sin(10 \cos^{-1} x) \frac{d}{dx} (10 \cos^{-1} x) \\ &= -\sin(10 \cos^{-1} x) \times 10 \times \frac{-1}{\sqrt{1-x^2}} \quad \left[\text{Since } \frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}} \right] \\ \text{or, } y_1 &= 10 \sin(10 \cos^{-1} x) \frac{1}{\sqrt{1-x^2}} \end{aligned}$$

$$\text{or, } \sqrt{1-x^2} y_1 = 10 \sin(10 \cos^{-1} x)$$

$$\text{or, } (1-x^2)y_1^2 = 100 \sin^2(10 \cos^{-1} x) = 100\{1 - \cos^2(10 \cos^{-1} x)\}$$

$$\therefore (1-x^2)y_1^2 = 100(1-y^2) \quad [\text{From equation <1>}]$$

Again, differentiating w. r. t. x we get,

$$(1-x^2) \frac{d}{dx} (y_1^2) + y_1^2 \frac{d}{dx} (1-x^2) = 100 \frac{d}{dx} (1-y^2) \quad \left[\text{Since } \frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx} \right]$$

$$\text{or, } (1-x^2)2y_1y_2 + y_1^2(0-2x) = 100(0-2yy_1)$$

$$\text{or, } (1-x^2)2y_1y_2 - 2xy_1^2 = -200yy_1$$

$$\therefore (1-x^2)y_2 - xy_1 = -100y \quad \dots \dots \dots <2>$$

Differentiating <2> 10 times w. r. t. x with the help of Leibnitz's theorem,

For first part, $(1-x^2)y_2$

Let $u = y_2$, $v = 1-x^2$

$$\begin{array}{l|l} \therefore u_1 = y_3 & \text{and } v_1 = 0 - 2x \\ \therefore u_2 = y_4 & \therefore v_2 = -2 \\ \dots\dots\dots & \\ u_{10} = y_{10+2} = y_{12} & \end{array}$$

Therefore, by Leibnitz's Theorem

$$\begin{aligned}
 (uv)_n &= u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + u v_n \\
 \{(1-x^2)y_2\}_{10} &= y_{12}(1-x^2) + 10 \cdot y_{11}(-2x) + \frac{10 \cdot 9}{2!} y_{10} \cdot (-2) \\
 &= (1-x^2)y_{12} - 20xy_{11} - 90y_{10}
 \end{aligned}$$

For 2nd part, $-xy_1$.

Let $u = y_1$, $v = -x$

$$\begin{array}{l|l}
 \therefore u_1 = y_2 & \text{and } v_1 = -1 \\
 \therefore u_2 = y_3 & \therefore v_2 = 0 \\
 \dots\dots\dots & \\
 \text{Thus, } u_{10} = y_{10+1} &
 \end{array}$$

Therefore, by Leibnitz's Theorem

$$\begin{aligned}
 \{(-x)y_1\}_{10} &= y_{10+1}(-x) + 10 \cdot y_{10} \cdot (-1) + 0 \\
 &= -xy_{11} - 10y_{10}
 \end{aligned}$$

For last part, $\{-100y\}_{10} = -100y_{10}$

Thus, from equation <2> we can write,

$$(1-x^2)y_{12} - 20xy_{11} - 90y_{10} - xy_{11} - 10y_{10} = -100y_{10}$$

$$\text{or, } (1-x^2)y_{12} - 21xy_{11} - 100y_{10} + 100y_{10} = 0$$

$$\text{or, } (1-x^2)y_{12} - 21xy_{11} = 0$$

$$\therefore (1-x^2)y_{12} = 21xy_{11}$$

Hence showed

HW

Problem.13 If $y = \sin^{-1} x$, then show that

$$\text{i) } (1-x^2)y_2 - xy_1 = 0.$$

$$\text{ii) } (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0.$$

Problem.14 If $y = \tan^{-1} x$, then show that

$$\text{i) } (1+x^2)y_1 = 1.$$

$$\text{ii) } (1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0.$$

Problem.15 If $\log y = \tan^{-1} x$, then show that

$$\text{i) } (1+x^2)y_2 + (2x-1)y_1 = 0.$$

$$\text{ii) } (1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + n(n+1)y_n = 0.$$

Problem.16 If $y = (\sin^{-1} x)^2$, then show that

$$\text{i) } (1-x^2)y_2 - xy_1 - 2 = 0.$$

$$\text{ii) } (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0.$$